Toward a four-dimensional Yang–Mills construction

Notebook 25: a smaller representative of the classical gauge orbit

Research working document

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Abstract

The contour-gauge stationary representative need not contract link angles under blocking. Before applying that rooted current gauge, however, the partial ordinary average of the established fine classical minimizer has link-angle bound C/R, where $R = L^s$ is the remaining averaging ratio. We prove the same factor in its strict rooted support norm and quadratic remainder, with constants uniform in both averaging depths at fixed base L. The proof applies Banach-valued local analytic estimates to complete output fibers and does not sum over their increasing number of coordinates. For an actual one-step gauge-invariant total correction, this representative supplies an analytic continuation of its stationary-composition germ wherever the smaller representative lies in the original domain. The support-coefficient norm and complete inherited fluctuation integral still require their own domain estimates.

This result addresses the angle-domain obstruction identified in Notebook 22. It uses the ordinary trajectory bound, the explicit ordinary minimizer, its localized construction, and the local gauge identity. At one physical step the displayed amplitude factor is C/L; the constant may depend on that fixed base L. The uniform C/L^s statement for larger s concerns the iterated constraint, and does not identify a new direct-block integration.

1 A small representative before the rooted current gauge

Fix the base odd ratio L, dimension four, $0 < \alpha < 1$, the compact matrix gauge group and compatible finite periodic geometry. Write

$$M = L^k, \qquad R = L^s, \qquad N = MR, \qquad \eta = N^{-1}, \qquad h_t = L^t\eta, \qquad h_k = R^{-1}. \tag{1}$$

Coordinates in this section have final next-coarse spacing one. The current lattice after k averages has spacing $1/R$. The datum $W = e^{iw}$ is a group datum, and w is its link angle. All constants below are uniform in k, s and volume for the fixed base L.

Let $A_{\mathrm{c},N}(w)$ be the established corrected fixed-gauge minimizer, and let T_N be the explicit ordinary-representative transfer. Thus

$$\begin{aligned} A_{\mathrm{o},N}(w) &= T_N(A_{\mathrm{c},N}(w)), \qquad U_N(w) = e^{i\eta A_{\mathrm{o},N}(w)}, \\ M_L^{k+s}(U_N(w)) &= e^{iw}, \qquad \|A_{\mathrm{o},N}(w)\|_{\mathcal{N}_\alpha} \le C_{\mathrm{o}}\|w\|_\infty. \end{aligned} \tag{2}$$

The last bound and a common complex product domain are uniform in the total depth $k+s$. Superscripts on M_L mean iteration of the fixed-L ordinary average. Define

$$V_{\mathrm{p},k,s}(w) = M_L^k(U_N(w)), \qquad a_{\mathrm{p},k,s}(w) = \frac{1}{i}\log V_{\mathrm{p},k,s}(w). \tag{3}$$

Exact composition gives

$$M_L^s(V_{\mathrm{p},k,s}(w)) = e^{iw}. \tag{4}$$

For $s = 1$ this is the pre-root-gauge representative used in the physical one-step construction. For $s > 1$ it is a representative of the iterated constraint (4); a separately defined direct R-block average, pivot chart and centered integral are not identified with that iteration here.

1.1 The exact amplitude factor

The proved ordinary-reference trajectory estimate is uniform in every prefix of the averaging sequence. With $U = e^{i\eta A}$ and $a = \|A\|_\infty$ sufficiently small, it states

$$\left\| \frac{1}{i} \log M_L^t(U) \right\|_\infty \le 3K_V h_t a, \qquad K_V = 2 + \frac{2K_\Gamma}{L-1}, \qquad K_\Gamma = 8m_\Gamma, \tag{5}$$

where m_Γ is a fixed one-step contour-length bound. In that proof the only accumulated scale sum is

$$\sum_{q<t} h_q = \frac{h_t - \eta}{L-1}. \tag{6}$$

In particular its constants depend on the base L, not on the endpoint ratio R. Apply (5) with $t = k$ and the ordinary minimizer in (2). On a smaller fixed ball,

$$\boxed{\|a_{\mathrm{p},k,s}(w)\|_\infty \le \frac{C_0}{R}\|w\|_\infty, \qquad C_0 = 3K_V C_{\mathrm{o}}.} \tag{7}$$

This is also a complex estimate on the common logarithm branch. When $k = 0$, it reduces to $a_{\mathrm{p},0,s} = \eta A_{\mathrm{o},N}$ with $\eta = 1/R$, checking the normalization.

The same trajectory proof gives a uniformly holomorphic physical map

$$P_{k,N}(A) = \frac{1}{ih_k} \log M_L^k(e^{i\eta A}), \qquad \|P_{k,N}(A)\|_\infty \le 3K_V \|A\|_\infty. \tag{8}$$

Each output reads a fine-field neighborhood of physical radius at most $c_L h_k$: the one-step dependence ranges sum geometrically. No physical adjoint of this ordinary map is used in this argument.

1.2 Strict supports without counting the output fiber

Support labels denote final unit cells, carrying the variables w. Let τ be periodic graph Steiner length, allowing auxiliary vertices. Fix an admissible exponent κ from the classical support theorem, and any fixed cardinality reserve $a \ge 0$. For a current lattice of spacing $h = 1/R$, collect all bond components whose initial sites lie in one final cell i into a Banach fiber $E_{h,i}$ with its maximum norm. The fiber dimension is of order R^d, but the norm is not a sum over those coordinates. For a strict support family define

$$\|u\|_{\kappa,a;r} = \sup_i \sum_{\varnothing \ne X} e^{\kappa\tau(\{i\}\cup X)+a|X|} \sup_{\|w_X\|_\infty < r} \|u_{i,X}(w_X)\|_{E_{h,i}}. \tag{9}$$

The output maximum is inside the support sum. Complete fine-field input families use the established smooth partition and complete $\mathcal{N}_\alpha$ cell spaces.

Theorem 1.1 (Supported pre-gauge representative)**.** *There are $r_a > 0$ and $C_a < \infty$, independent of k, s and volume, such that for $0 < r \le r_a$ the representative (3) admits absolutely convergent strict holomorphic families*

$$a_{\mathrm{p},k,s}(w)|_i = \sum_{\varnothing \ne X} a_{\mathrm{p};i,X}(w_X), \qquad a_{\mathrm{p};i,X}(0) = 0, \qquad \|a_{\mathrm{p},k,s}\|_{\kappa,a;r} \le \frac{C_a r}{R}. \tag{10}$$

There is a uniformly supported linear operator $H_{\mathrm{p},k,s} = D(P_{k,N}T_N)(0)H_N$ such that

$$\|a_{\mathrm{p},k,s} - R^{-1}H_{\mathrm{p},k,s}w\|_{\kappa,a;r} \le \frac{C_a r^2}{R}. \tag{11}$$

Here H_N is the corrected flat minimizing extension. The selected linear operator is not identified with a separately rooted source gauge extension.

Proof. The classical support construction supplies $A_{\mathrm{c},N}(w)$ in complete fine-field cells with norm at most $C_{a,0}r$, uniformly in N and volume, including its linear part and quadratic remainder. The ordinary transfer T_N is uniformly holomorphic $\mathcal{N}_\alpha \to \mathcal{N}_\alpha$ on a fixed ball, vanishes at zero, and has fixed final-physical dependence radius. Indeed its cumulative endpoint gauge is a local bounded analytic function of the fine potential, its point extension is a fixed smooth local interpolation, and its fine gauge action is a local analytic jet map with the apparent inverse-mesh singularity removed before taking derivatives.

Consequently $F_{k,N} = P_{k,N} \circ T_N$ is uniformly bounded and holomorphic on a fixed $\mathcal{N}_\alpha$ ball into the physical bounded output field, and $F_{k,N}(0) = 0$. Its dependence radius in final units is fixed. For the entire fiber $E_{h,i}$ the union of the individual output neighborhoods lies in one fixed collar of i, even though the number of components inside the fiber grows with R.

Let ρ be a common analytic input radius, C_{rec} the smooth reconstruction norm, D_0 the fixed number of input cell labels in that collar, and R_0 its graph radius. Banach-valued Cauchy and polarization give for the symmetric Taylor coefficients, including the factor $1/n!$, the mixed cell-slot bound

$$v_n \le Kc^n, \qquad c = \frac{eC_{\mathrm{rec}}D_0 e^{\kappa R_0}}{\rho}, \qquad n \ge 1. \tag{12}$$

The constant K bounds the output in its maximum norm on the analytic ball. Cauchy is applied to that whole Banach output; it is not applied to each scalar component followed by a sum over R^d components. The coefficient norm places the output maximum before the sum over input cell labels, precisely as required by (9).

For substituted input supports $X_1, \dots, X_n$ rooted at $j_1, \dots, j_n$, the exact gluing inequality is

$$\tau\left(\{i\} \cup \bigcup_q X_q\right) \le \tau(\{i, j_1, \dots, j_n\}) + \sum_q \tau(\{j_q\} \cup X_q). \tag{13}$$

Also $e^{a|\cup_q X_q|} \le \prod_q e^{a|X_q|}$. The fixed coefficients of the local map add no variable support. Thus substitution of an input family of norm t has output norm at most

$$\sum_{n\ge1} Kc^n t^n = \frac{Kct}{1-ct} \le 2Kct \quad \text{if } ct \le \tfrac{1}{2}. \tag{14}$$

Choose r_a so $cC_{a,0}r_a \le 1/2$, and shrink it to respect every previous evaluated-branch domain. These choices are independent of R. Finally $a_{\mathrm{p},k,s} = h_k F_{k,N}(A_{\mathrm{c},N}(w))$ gives (10). Separating the linear terms in the two analytic series gives (11). Absolute convergence justifies rearrangement and identifies the sum with the actual representative. □

This proof neither uses the nonuniform ordinary-average physical adjoint nor infers tree locality from a one-weight response bound. The spatial exponent is in final unit cells; in current-unit distances it has length scale R. Uniformity in $R = L^s$ keeps the base L fixed.

1.3 One-step physical identification and analytic continuation

Set $s = 1$. The physical construction uses the actual averaged-contour current gauge

$$u(y) = 1, \qquad u(x) = P_{V_{\mathrm{p}}}(y, x) \quad (x \text{ in the block of } y), \qquad V_*(W) = V_{\mathrm{p}}(W)^u. \tag{15}$$

It has $P_{V_*} = 1$, $M_L(V_*) = W$ and, on the common complex branch,

$$\| \log u/i \|_\infty \le C_L \|a_{\mathrm{p}}\|_\infty \le C'_L \|w\|_\infty. \tag{16}$$

The classical local gauge-invariance theorem applies to both configurations and this gauge path after a common fixed shrink. Hence $S_M(V_*) = S_M(V_{\mathrm{p}})$. The existing minimum comparisons and common strict-convexity uniqueness identify (15) with the physical centered branch. Gauge-zero is obtained by the explicit gauge and that comparison, not from stationarity alone. The gauge in (16) need not have size w/L; boundary angles of V_* can be of order w even when (7) holds. Thus the earlier pure-gauge obstruction for the rooted representative is preserved.

Let E be a total holomorphic current-field functional on the angle product ball $\|a\|_\infty < b$, locally gauge invariant when its two arguments and the gauge lie in their common invariance domains. Then

$$E_{\mathrm{p}}(w) = E(a_{\mathrm{p},k,1}(w)) \tag{17}$$

is holomorphic for

$$\|w\|_\infty < \min\{r_{\mathrm{class}}, Lb/C_0\}, \tag{18}$$

with smaller margins if required. On a sufficiently small overlap around zero, both original compositions and the gauge path are admissible, so

$$E(a_*(w)) = E(a_{\mathrm{p},k,1}(w)). \tag{19}$$

The real gauge identity and holomorphic uniqueness prove this equality. Therefore (17) is an analytic continuation of the original stationary-composition germ to (18). It does not assert that E was already defined at $a_*(w)$ outside its original ball. No coefficientwise identity for an arbitrary decomposition $E = \sum_H E_H$ is inferred.

If E is an inherited logarithm, its original domain must already carry a nonvanishing integral and compatible holomorphic logarithm. The covariance of a possibly vanishing integral alone is insufficient. For $s > 1$, $E(a_{\mathrm{p},k,s})$ remains an analytic composition with the same amplitude improvement, but identifying it with a particular macrostep stationary germ requires that macrostep's branch and gauge comparison. A direct-R chart is not supplied here.

For any fixed $R = L^s > C_0$, the representative itself maps an angle ball of radius b into the strictly smaller current angle ball of radius $C_0 b/R$, provided b is below the uniform classical radius. This geometric domain inclusion is proved; invariance is needed to identify the resulting composition with a stationary germ, not for the inclusion itself. It is not an activity-norm contraction.

1.4 The activity norm still has a coordinate cost

Suppose the inherited connected-carrier activities have scalar coordinate polydisc radius b, at most D coordinates per actual old cell, and

$$\sup_i \sum_{H \ni i} e^{a_0|H|} \|E_H\|_b \le \epsilon. \tag{20}$$

Let π project those old cells to final cells, preserving connected images, with fibers bounded by F_R. Put $\beta = a_{\mathrm{out}} + \mu$. The established carrier-transport theorem, applied to (10), gives output norm at most $F_R S_\delta \epsilon$ whenever

$$\beta + \delta \le \kappa, \qquad \beta + D \log \frac{1}{1 - C_a r/(Rb)} \le a_0. \tag{21}$$

Here D is the actual old-cell scalar-coordinate count, not the output Banach fiber dimension in Theorem 1.1. Nevertheless if the chosen background/Gaussian cell has $D \asymp R^d$, the small-radius sufficient condition in (21) still has $r/b = O(R/D) = O(R^{1-d})$. If D stays fixed, that particular loss disappears but the projection factor F_R can grow like R^d. Analytic-domain inclusion and a contracting activity-norm induction are distinct assertions.

There is a further pointwise identity on the original common domain: with $V_* = V_{\mathrm{p}}^u$,

$$(e^{iB}V_*)^{u^{-1}}(b) = e^{i\,\mathrm{Ad}_{u(b_-)^{-1}}\,B(b)}V_{\mathrm{p}}(b). \tag{22}$$

For real u the adjoint preserves the invariant color norm; it is uniformly bounded on the small complex domain. Using the right side to extend an inherited insertion with $B = \Phi_{V_*}(gz)$ is a possible next step. The coupling is inside the nonlinear chart; generally $\Phi_V(gz) \neq g\Phi_V(z)$. This section does not prove a larger-domain normalized Gaussian-integral identity. That needs the actual chart and fluctuation bounds, holomorphic domination, interaction norm control and equality of the two analytic integrals on an overlap, with the exact Gaussian, gauge penalty, pivot/Haar factors and cutoff arguments retained. No stable quantum iteration or running-coupling conclusion is made.

References

[1] *Toward a four-dimensional Yang–Mills construction*, research Notebooks 14–16 and 19–24 in this workspace. The ordinary reference trajectory, uniform classical minimum, ordinary representative transfer and strict support calculus are the principal inputs.

[2] T. Bałaban, *Renormalization group approach to lattice gauge field theories I*, Commun. Math. Phys. **109** (1987), 249–301, original article copy.

[3] E. Sussman and collaborators, Yang–Mills-LLM-experiment, deposited raw accounts at commit `e392ef45013a581664bab76b90867170b30fe10e`.